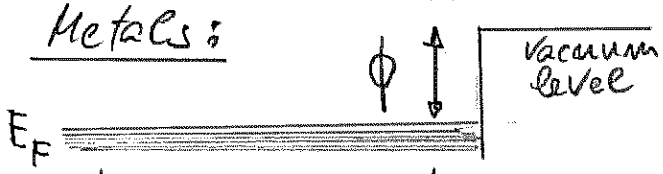
Quantum detectors → Photoemissive detectors

Photocathode materials

Photoelectric effect

Metals:



ϕ = work function

$$h\nu > e\phi \Rightarrow \lambda < \frac{hc}{e\phi}$$

$$E_{K_{max}} = h\nu - e\phi$$

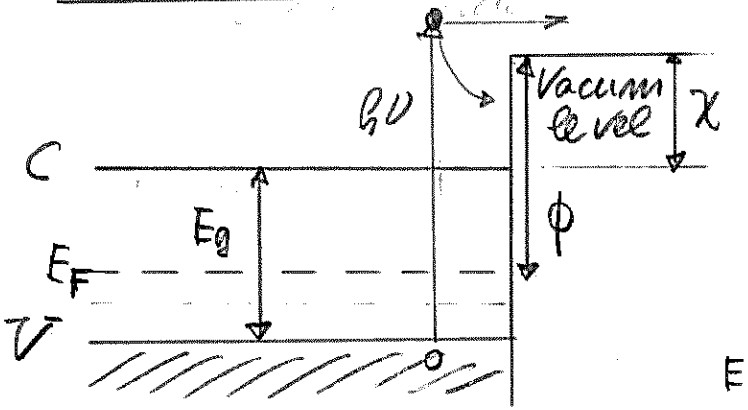
low λ threshold

Low quantum efficiency due to electron collisions

$\phi_{Cs} = 1.95\,V \rightarrow \lambda_{th} \simeq 600\,nm$

$\eta_{Cs} \simeq 0{,}1\,\%$

Semiconductors



$$h\nu > e(E_g + \chi)$$

$$\lambda < \frac{hc}{e(E_g+\chi)}$$

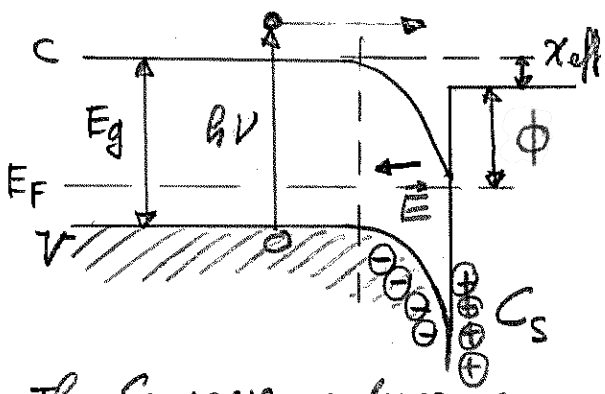
fast electrons collide with lattice ions and exchange phonons → low energy loss

Photocathodes based on Antimony + Alkalin Metals

- Bialkali: Na_2KSb $\lambda_{th} = 650\,nm$ $\eta = 20\%$
- (S20) Multialkali: $NaKCsSb$ $\lambda_{th} = 850\,nm$ $\eta = 20\%$
- Ext Red MA (S25) $\lambda_{th} = 900\,nm$ $\eta = 10\%$
- Ag Cs O (S1) $\lambda_{th} = 1200\,nm$

NEA

P-doped semiconductor (GaAs) activated with Cs on Cs_2O



The Cs ions induce a depletion layer of neg. charge → E field → bending of the bands

χ becomes negative

$$h\nu > E_g \rightarrow \lambda_{th} = \frac{hc}{eE_g}$$

electrons can escape even also thermolization

GaAs:CsO $\lambda_{th} = 900\,nm$

$\eta \simeq 25\,\%$

GaAsP → $\lambda_{th} = 750\,nm$

$\eta \simeq 50\,\%$

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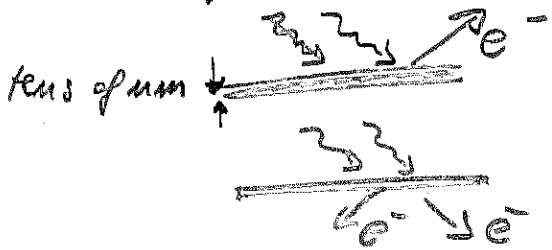
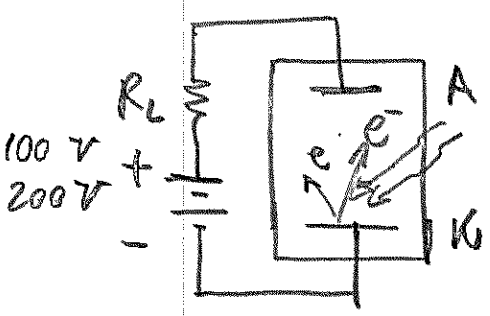
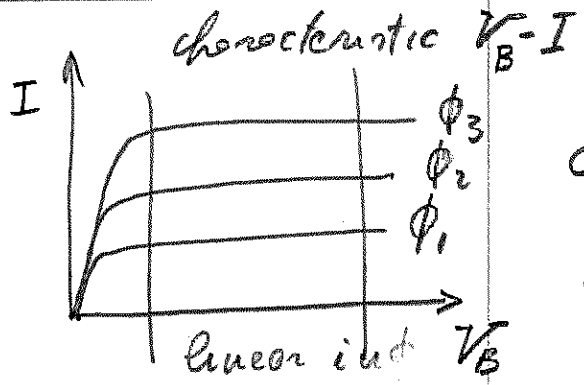
Photocathodes

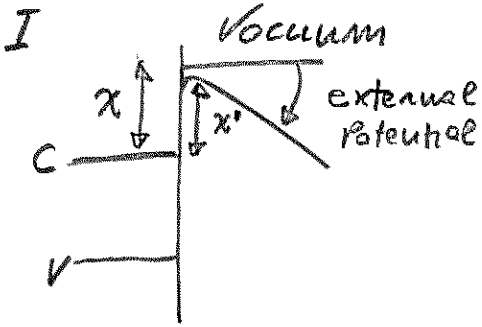
- Thin layer of semiconductors deposited on glass (borosilicate) or quartz (SiO_2) or UV transparent materials (MgF_2)
- Thickness of few tens of nanometers $\left[I = I_0 \exp(-\mu_q x) \right]$
- Electrons can be emitted from the same side of incident photons or from the opposite side



Vacuum Photodiodes







$$I_\lambda = \phi_\lambda \eta e = \frac{P_\lambda \eta e \lambda}{hc} \qquad \phi_\lambda = \frac{P_\lambda \lambda}{hc}$$

$$R_\lambda^{(I)} = \frac{I_\lambda}{P_\lambda} = \eta \frac{e \lambda}{hc} \qquad R_\lambda^{(V)} = R_\lambda^{(I)} \cdot R_L \qquad \frac{e}{hc} = 0{,}8 \cdot 10^{+6} \rightarrow R_\lambda \simeq 0{,}1 \ A/W$$

Poor sensitivity. Replaced by silicon PD unless for detecting UV radiation.

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Noise in vacuum photodiodes

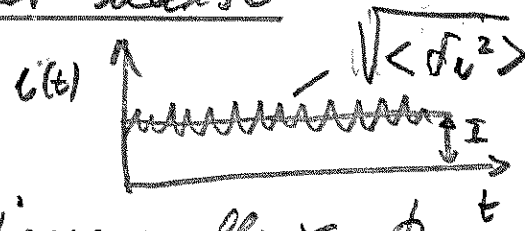
Thermionic emission (Richardson eq.) $\longrightarrow$ dark current

$$J_T = \underbrace{\left[\frac{k^2 m_e e}{2\pi^2 \hbar^3}\right]}_{\alpha} T^2 \exp\left(-\frac{e\phi}{kT}\right) = \alpha T^2 \exp\left(-\frac{e\phi}{kT}\right) \qquad \alpha \simeq 1{,}2\cdot 10^6\ A/m^2 K^{-2}$$

@ 300 K $\quad J_T \simeq 10^{-15} - 10^{-11}\ A/m^2 \quad$ Cooling reduces J_T

other effects cause cathode emissions $\begin{cases} \text{ionic feedback} \\ \text{cosmic rays} \end{cases}$

Shot noise



Signal only

stationary flux ϕ_s

$$I_s = e\eta\phi_s = \eta\frac{e}{hc}\lambda P_s$$

$$R_\lambda = \eta\frac{e\lambda}{hc}$$

$$i_{shot}^{RMS} = \sqrt{\langle \delta i^2 \rangle} = \sqrt{\langle [i(t) - I_s]^2 \rangle} = \sqrt{2 e I_s \Delta f}$$

$$SNR = \sqrt{\frac{I_s^2}{\langle \delta i^2 \rangle}} = \sqrt{\frac{I_s^2}{2 e I_s \Delta f}} = \sqrt{\frac{I_s}{2e\Delta f}} = \sqrt{\frac{P_s \eta \lambda}{2hc\Delta f}}$$

SNR : $\uparrow \sqrt{\lambda} \quad \downarrow \sqrt{\Delta f}$

$$SNR = 1 \longrightarrow I_{min} = 2e\Delta f \qquad P_{min} = \frac{2hc\Delta f}{\eta\lambda} = 2h\nu\frac{\Delta f}{\eta} \quad \begin{matrix}\Delta f = 1 \\ \eta = 0{,}5\end{matrix} \begin{cases} 2\ e/sec \\ 4\ Ph/sec \end{cases}$$

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NEP of a Vacum photodiode

$I = I_d \rightarrow$ Thermionic emission

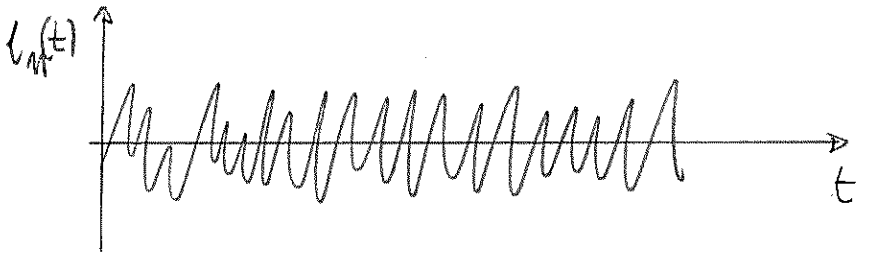
$$i_s^{RMS} = \sqrt{2e I_d \Delta f} \qquad i_{Th}^{RMS} = \sqrt{\frac{4KT}{R_L}\Delta f}$$

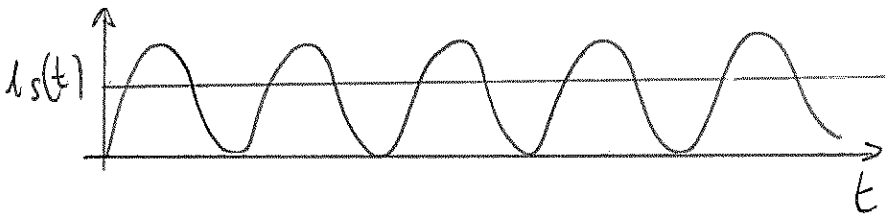
$$i_{noise}^{RMS} = \sqrt{\left[2e I_d + \frac{4KT}{R_L}\right]\Delta f}$$

$$\simeq 10^{-38}\ A^2 \cdot s \qquad \simeq 10^{-22}\ A^2 \cdot s$$

$$NEP = \frac{i_{Th}^{RMS}}{R_\lambda} = \frac{hc}{e\eta\lambda}\sqrt{\frac{4KT}{R_L}\Delta f}$$

$NEP \simeq \underline{1\ \mu W}$ very high value





$$\left(i_n^{RMS}\right)^2 = \left(i_s^{RMS}\right)^2 + \left(i_T^{RMS}\right)^2$$

sum of the variances for indipent Processes

$$R = \frac{\eta e \lambda}{hc}$$

$I_d \simeq 10^{-11}\ A$ $\quad \left[\begin{array}{l} T = 300\ K \\ A \simeq \text{Few } cm^2 \end{array}\right]$

$R_L = 50\ \Omega$

$KT \simeq 25\ meV \simeq 4 \cdot 10^{-21}\ J$

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